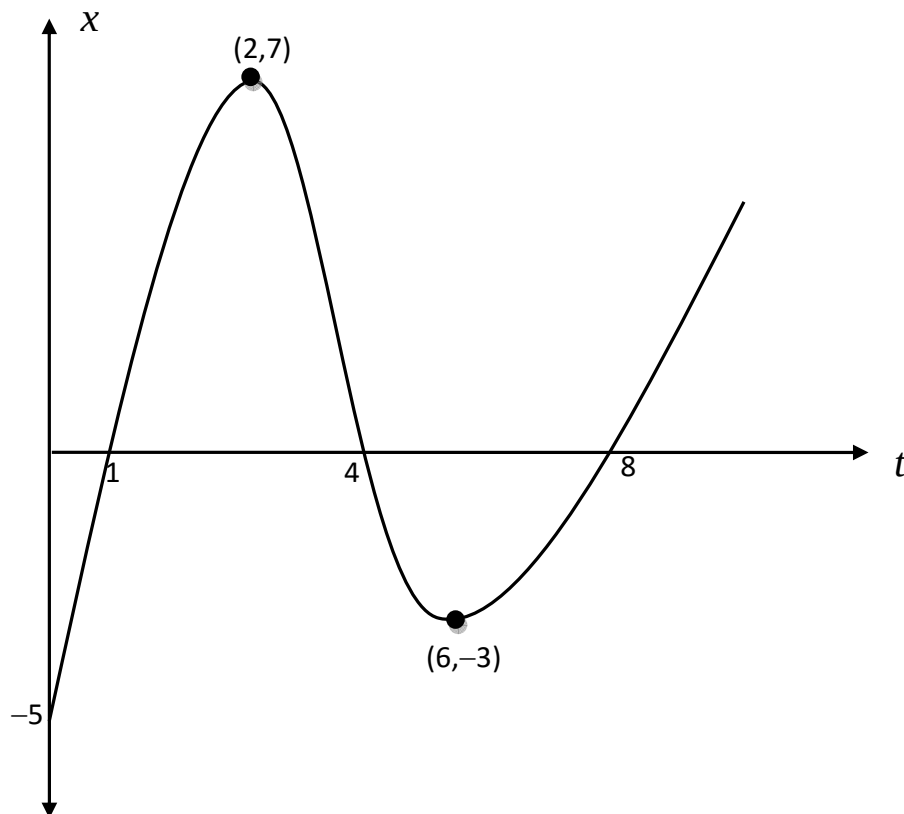


Question 1 (10 marks)**Marks**

- a) The graph shows displacement, x metres from the origin, at any time t seconds, of a particle moving in a straight line. **5**



- Where was the particle initially?
- When was the particle at the origin?
- When was the particle at rest?
- Estimate the time when the acceleration was zero.
- How far did the particle travel in the first 8 seconds.

Question 1 continued.**Marks**

- b) The acceleration of a particle, at time t seconds, is given by $\frac{d^2x}{dt^2} = 6 + \frac{2}{t^2} \text{ cm/s}^2$. **5**

When $t = 1$, $\frac{dx}{dt} = -3$ and $x = 2$. Find correct to 4 significant figures, the value of x when $t = 2$.

Question 2 (10 marks) Start this question on a new page.

- a) A full water tank is being drained and the number of litres in the tank, L , at time t minutes is given by the equation $L = 100(50 - t)^2$. **4**
- At what rate is the water draining out of the tank when $t = 6$ minutes?
 - What is the capacity of the tank?
 - How long will it take to completely empty the tank?
- b) The population of rabbits, N , on an island was estimated to satisfy the equation $N = N_0 e^{kt}$ where N_0 and k are constants and t is measured in months. **6**
- Initially the rabbit population was 500 and at the end of 8 months there were 2200 rabbits. Find the value of k to 3 decimal places.
 - Find the number of rabbits on the island after 1 year. (Give your answer correct to the nearest hundred).
 - How long will it take for the rabbit population to exceed 20 000?
 - Show that $\frac{dN}{dt} = kN$.

Question 3 (10 marks) Start this question on a new page.**Marks**

- a) The third term of an arithmetic series is 7 and the tenth term is 42. **3**
- i) Find the first term and the common difference.
 - ii) Find the sum of the first 12 terms.
- b) The limiting sum of a geometric series is 4 and the first term is 1. **3**
- i) Determine the common ratio
 - ii) Determine the fourth term of the sequence in rational form.
- c) A person invests \$2 000 at the beginning of each year in a superannuation fund. **4**
Compound interest is paid at 8% per annum on the investment. The first \$2 000 is invested at the beginning of 2009 and the last is to be invested at the beginning of 2038. Calculate to the nearest dollar:
- i) the amount to which the 2009 investment will have grown by the beginning of 2039.
 - ii) the total value of the person's investment by the beginning of 2039.

Question 4 on page 4.

Question 4 (10 marks) Start this question on a new page.**Marks**

- a) A particle moves so that its distance x cm from a fixed point O at time t seconds is given by $x = 2 \cos 3t$. Determine when the particle first comes to rest after $t = 0$. **2**
- b) When Jack left school he borrowed \$25 000 to buy his first car. The interest rate on the loan was 9% per annum and Jack planned to pay back the loan in 60 equal monthly repayments of \$M. **8**
- i) Show that the amount owing immediately after making his first monthly repayment was $\$[25000 \times 1.0075 - M]$.
- ii) Show that the amount owing immediately after making his third monthly repayment was $\$[25000 \times 1.0075^3 - M(1 + 1.0075 + 1.0075^2)]$.
- iii) Calculate the value of M.
- iv) If Jack could afford to make monthly repayments of \$1000, how long would it take him to repay the loan?

End of paper.

Question 1

i) Particle initially 5 units left of origin ①

ii) At the origin when $t = 1, 4, 8$ ①

iii) At rest when $t = 2$ and $t = 6$ ①

iv) Acceleration zero when $t = 3$ ①

v) Distance = $5 + 7 + 7 + 3 + 3$
 $= 25\text{m}$ ①

b) $a = 6 + \frac{2}{t^2}$

$$\begin{aligned} v &= \int a \, dt \\ &= \int (6 + 2t^{-2}) \, dt \\ &= 6t - 2t^{-1} + c \quad ① \\ &= 6t - \frac{2}{t} + c \end{aligned}$$

When $t = 1$ $v = -3$

$$\therefore -3 = 6 \times 1 - \frac{2}{1} + c$$

$$-3 = 6 - 2 + c$$

$$\therefore c = -7 \quad ①$$

$$\therefore \frac{dx}{dt} = 6t - 2t^{-1} - 7$$

$$x = \int \frac{dx}{dt} \, dt$$

$$\begin{aligned} x &= \int (6t - 2t^{-1} - 7) \, dt \\ &= \int 6t - \frac{2}{t} - 7 \, dt \quad ① \\ &= 3t^2 - \frac{2}{t} - 7t + k \end{aligned}$$

$$x = 3t^2 - 2 \ln t - 7t + k$$

When $t = 1$ $x = 2$

$$\begin{aligned} \therefore 2 &= 3 \times 1^2 - 2 \ln 1 - 7 \times 1 + k \\ 2 &= 3 - 0 - 7 + k \quad ① \\ \therefore k &= 6 \end{aligned}$$

$$\therefore x = 3t^2 - 2 \ln t - 7t + 6$$

$$t = 2 \quad x = 2.614 \quad ①$$

QUESTION 2

a) $L = 100(50-t)^2$

Rate Implies $\frac{dL}{dt}$

$$\frac{dL}{dt} = 2 \times 100(50-t) \times -1$$

$$= -200(50-t) \quad (1)$$

When $t=6$ $\frac{dL}{dt} = -200(50-6)$

$$= -8800 \quad (1)$$

$\therefore$ Draining at a rate of 8800 L/Min

a) Capacity when $t=0$

$$\text{Capacity} = 100(50-0)^2$$

$$= 250000 \text{ L} \quad (1)$$

iii) Empty when $L=0$

$$100(50-t)^2 = 0$$

$$50-t = 0 \quad (1)$$

$$t = 50$$

$\therefore$ Empty after 50 minutes

b) $N = N_0 e^{kt}$

When $t=8$ $N=2200$, $N_0 = 500$

$$\therefore 2200 = 500 e^{k \times 8}$$

$$e^{8k} = \frac{2200}{500}$$

$$e^{8k} = 4.4$$

$$k = \frac{\ln 4.4}{8} \quad (2)$$

$$k = 0.1852 \dots (\text{Calc})$$

$$k = 0.185$$

ii) $N = N_0 e^{kt}$

$$= 500 e^{0.185 \times 12}$$

$$= 4603.6 \dots (\text{Calc})$$

Population after 1 year = 4600 rabbit

iii) First t when $N > 20000$

$$20000 = 500 e^{0.185t}$$

$$40 = e^{0.185t}$$

$$t = \frac{\ln 40}{0.185} \quad (2)$$

$$\therefore t > 19.93 \dots (\text{Calc})$$

$\therefore$ Population > 20000 after 20 months

iv) $N = N_0 e^{kt}$

$$\frac{dN}{dt} = k N_0 e^{kt}$$

Since

$$N = N_0 e^{kt}$$

$$\frac{dN}{dt} = kN$$

Q.E.D

$$Q3(a)(i) T_3 = 7 \quad T_{10} = 42.$$

$$\left. \begin{array}{l} a + 2d = 7. \quad (1) \\ a + 9d = 42 \quad (2) \end{array} \right\} \checkmark$$

$$(2) - (1) \quad 7d = 35$$

$$d = 5.$$

$$a = 7 - 10 = -3. \text{ First term } -3, \text{ common diff } 5. \quad \left. \right\} \text{ both } \checkmark (2).$$

$$(ii) S_{12} = \frac{12}{2} (2 \times -3 + 11 \times 5).$$

$$= 294. \quad (1) \checkmark$$

$$(b)(i) S_{\infty} = \frac{a}{1-r}$$

$$4 = \frac{1}{1-r} \quad \checkmark$$

$$4 - 4r = 1$$

$$4r = 3$$

$$r = \frac{3}{4}. \quad \checkmark \quad (2)$$

$$(ii) T_4 = 1 \times \left(\frac{3}{4}\right)^3$$

$$T_4 = \frac{27}{64}. \quad \checkmark \quad (1)$$

$$(c)(i) A_1 = 2000 \times 1.08^{30}$$

$$= 20125.31 \quad \checkmark$$

(1).

$$(ii) A_2 = 2000 \times 1.08^{29}$$

$$A_{30} = 2000 \times 1.08.$$

$$\text{Total} = A_1 + A_2 + \dots + A_{30}$$

$$= 2000 \times 1.08^{30} + 2000 \times 1.08^{29} + \dots + 2000 \times 1.08.$$

$$= 2000 (1.08^{30} + 1.08^{29} + \dots + 1.08)$$

$$\text{is GP } a = 1.08 \quad r = 1.08 \quad n = 30.$$

$$= 2000 \times \frac{1.08(1.08^{30} - 1)}{1.08 - 1} \quad \checkmark$$

$$= \$ 244592 \text{ (nearest dollar)} \quad \checkmark$$

$$(3) \quad (224566. (if 29))$$

Q4 a) $x = 2 \cos 3t$
 $\dot{x} = -6 \sin 3t$ ✓

$-6 \sin 3t = 0$
 $\sin 3t = 0$
 $3t = 0, \pi, \dots$
 $t = \pi/3$ ✓

b) i) $A_1 = 25000 + \frac{9/12}{100} \times 25000 - m$ ✓
 $= 25000(1.0075) - m$

ii) $A_2 = A_1(1.0075) - m$
 $= (25000(1.0075) - m)(1.0075) - m$
 $= 25000(1.0075)^2 - m(1.0075) - m$ ✓

$A_3 = A_2(1.0075) - m$
 $= (25000(1.0075)^2 - m(1.0075) - m)(1.0075) - m$
 $= 25000(1.0075)^3 - m(1.0075)^2 - m(1.0075) - m$
 $= 25000(1.0075)^3 - m(1 + (1.0075) + 1.0075^2)$

iii) $A_{60} = 0$
 $25000(1.0075)^{60} = m(1 + (1.0075) + \dots + 1.0075^{59})$
 $m = \frac{25000(1.0075)^{60}}{\frac{1(1.0075^{60} - 1)}{1.0075 - 1}}$ ✓
 $= \$518.96$ ✓

iv) $25000(1.0075)^n = 1000(1 + (1.0075) + \dots + 1.0075^{n-1})$
 $= 1000 \left(\frac{1(1.0075^n - 1)}{1.0075 - 1} \right)$ ✓
 $108333.33(1.0075)^n = 133333.33$
 $1.0075^n = 1.23076923$ | $n = \frac{\log 1.23076923}{\log 1.0075}$
 $n = 27.8$ m ✓